

Transcriptomic effect of miR-551b inhibition in acute myeloid leukaemia cells, and the behaviour of a ciliary/centrosomal gene set and a curated interaction network

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[Affiliation] · Analysis performed end-to-end with the open transcriptomics pipeline

Abstract. Cilia and centrosomal genes have been proposed as candidate markers in cancer, and miR-551b is an oncogenic microRNA in several solid tumours. We tested whether inhibiting miR-551b reshapes the transcriptome of acute myeloid leukaemia (AML) cells and, in particular, whether it perturbs a curated 683-gene cilia/ciliopathy set or a 25-gene STRING-style interaction network (dopaminergic, innate-immune, autophagy, iron and ciliary genes). Eight single-end RNA-seq runs (BioProject PRJNA1513000; primary AML blasts and the KG1a cell line, each treated with LNA or ZIP anti-miR against miR-551b versus a control oligonucleotide) were processed from raw reads: multi-connection download from the SRA AWS Open-Data mirror, Salmon selective-alignment quantification against GENCODE v44 (72-83% mapping), and DESeq2 differential expression. Using a paired, block design (adjusting for cell type and antisense chemistry), only three genes were significant (FDR<0.05; all down-regulated: TCF7, IGKC and one unannotated locus). The cilia set showed no significant gene (0/563) and was in fact less variable than the transcriptome background (mean $|\log_2\text{FC}|$ 0.185 vs 0.321); the interaction network showed no significant gene (0/17 expressed; the eight dopaminergic members were not detectably expressed in AML). Analysed separately, primary AML samples yielded 33 significant genes dominated by T- and B-lymphocyte markers (TRBC1, ZAP70, LCK, IGKC, IGHM) - a signature attributable to differing immune-cell content between samples rather than to miR-551b - while the clean KG1a line yielded essentially none. We conclude that, in this dataset, miR-551b inhibition produces a minimal and non-reproducible transcriptional footprint in AML cells and does not engage ciliary/centrosomal genes. This is a negative result and should be read against a clear power limitation: the design provides only one biological unit per unique condition, with the two chemistries used as pseudo-replicates.

1 Introduction

Primary cilia and the centrosomal/basal-body apparatus coordinate signalling (Hedgehog, Wnt, PDGFR) and cell-cycle progression, and their gene programme has been proposed as a candidate axis in tumour biology. Whether this programme is relevant to acute myeloid leukaemia (AML) is far from obvious: haematopoietic cells are largely non-ciliated, so any signal is more plausibly centrosomal than motile-ciliary. miR-551b, in turn, is an oncogenic microRNA amplified in ovarian and gastric cancers; antisense inhibition of miR-551b has been used to probe its downstream programme.

Here we ask a focused question: does inhibiting miR-551b in AML cells change the expression of (i) a curated 683-gene cilia/ciliopathy set and (ii) a 25-gene interaction network (assembled from a STRING-style neighbourhood spanning dopaminergic signalling, innate immunity, autophagy, iron handling and two core ciliary genes, IFT88 and ARL13B)? We processed the data end-to-end from raw reads and treat any result as hypothesis-level, given the study's limited replication.

2 Methods

2.1 Data

Eight single-end Illumina HiSeq 2000 RNA-seq runs were obtained from NCBI BioProject PRJNA1513000 (runs SRR40166669-SRR40166676). The design is a balanced 2x2x2: cell context (primary AML blasts vs the KG1a AML cell line) x antisense chemistry (LNA vs ZIP) x treatment (anti-miR-551b vs control oligonucleotide). No processed count matrix was deposited (no linked GEO series), so raw reads were used. Because the archive throttles single-stream transfers, reads were fetched at full depth (13-25 million reads/run) from the SRA AWS Open-Data mirror with a 16-connection downloader.

2.2 Quantification

Reads were quantified with Salmon 2.5.1 in selective-alignment mode against a GENCODE v44 transcriptome index (251,955 transcripts); transcript estimates were summed to gene level (tximport-style). Mapping rates were 71.8-72.7% for primary-AML samples and 80.7-82.9% for KG1a, consistent with cleaner cell-line input. The gene x sample matrix contained 60,883 genes; 25,994 passed a total-count ≥ 10 filter and were tested.

2.3 Differential expression and gene sets

Differential expression used DESeq2 (pyDESeq2). The primary model was a paired/blocked design, $\sim \text{cell_type} + \text{chemistry} + \text{treatment}$, testing anti-miR-551b vs control and thereby removing both the cell-context and chemistry axes; secondary per-cell-type models ($\sim \text{chemistry} + \text{treatment}$; $n=2$ vs 2) were also fit. Significance was Benjamini-Hochberg $\text{FDR} < 0.05$. Two gene sets were intersected with the results by gene symbol: a user-supplied 683-gene cilia/ciliopathy set and the 25-gene interaction network. Software: Python 3.12, Salmon, pyDESeq2, gseapy, pandas, scikit-learn, matplotlib.

3 Results

3.1 Sample structure

Principal-component analysis (Figure 1) is dominated by cell context: primary-AML and KG1a samples separate completely, which is why cell type is a blocking term in the model. Within each block the anti-miR-551b and control samples do not form a clear separate axis.

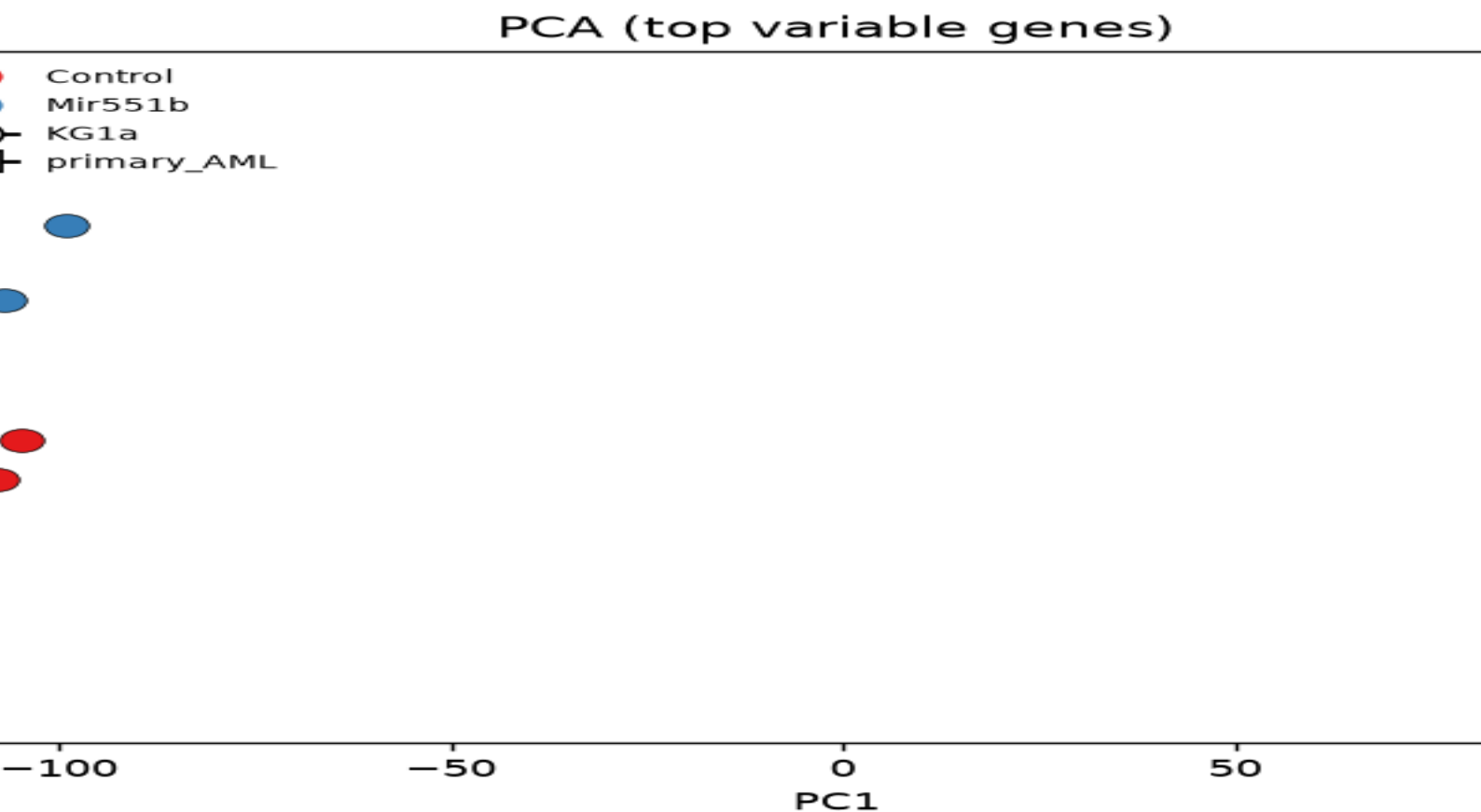


Figure 1. PCA of the eight samples (DESeq2-normalised, log-scaled counts). Colour = treatment (anti-miR-551b vs control); the dominant split is cell context (primary AML vs KG1a).

3.2 Genome-wide differential expression (paired model)

Adjusting for cell type and chemistry, only three genes were significant at $FDR < 0.05$, all down-regulated upon miR-551b inhibition (Table 1, Figure 2). The strongest annotated hits were TCF7 (a Wnt/T-cell transcription factor) and IGKC (an immunoglobulin constant region); the third is an unannotated locus. This is a near-null genome-wide response.

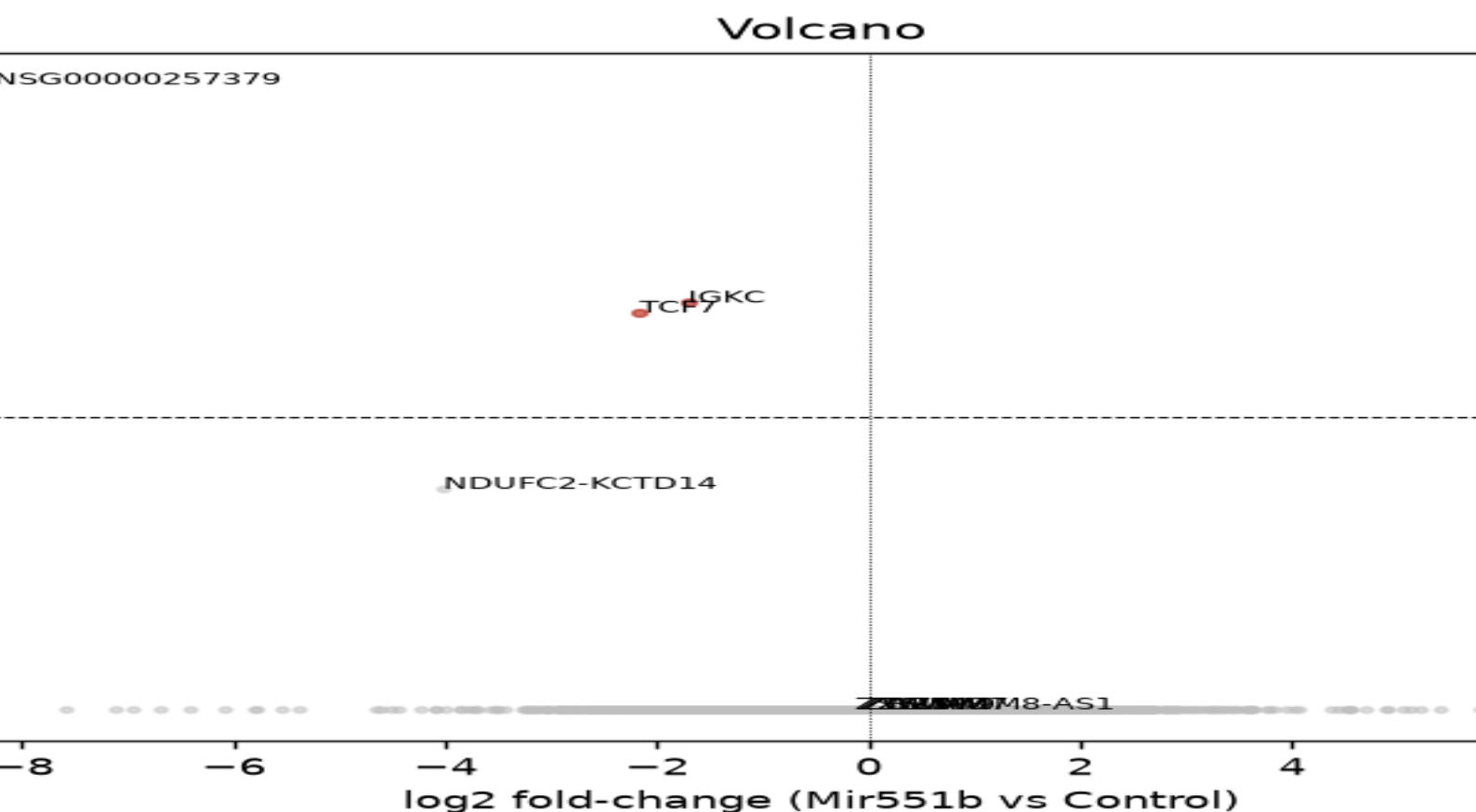


Figure 2. Volcano plot, anti-miR-551b vs control (paired model). Red points are the three FDR-significant genes; the bulk of the transcriptome is unchanged.

Table 1. All genome-wide significant genes (paired model, $FDR < 0.05$).

Gene	log2FC	Adj. p
ENSG00000257379	-8.46	1.6e-3
IGKC	-1.71	1.5e-2
TCF7	-2.16	1.7e-2

3.3 The cilia/centrosomal set does not respond

Of the 683-gene cilia set, 563 genes were expressed and testable; none reached $FDR < 0.05$ (Figure 3). Beyond simple non-significance, cilia-set genes were less variable than the transcriptome as a whole (mean absolute log2 fold-change 0.185 vs 0.321 background) - i.e. they are among the more stable genes under miR-551b inhibition. The most-shifted cilia genes by nominal p (Table 2) are unremarkable and do not survive multiple-testing correction.

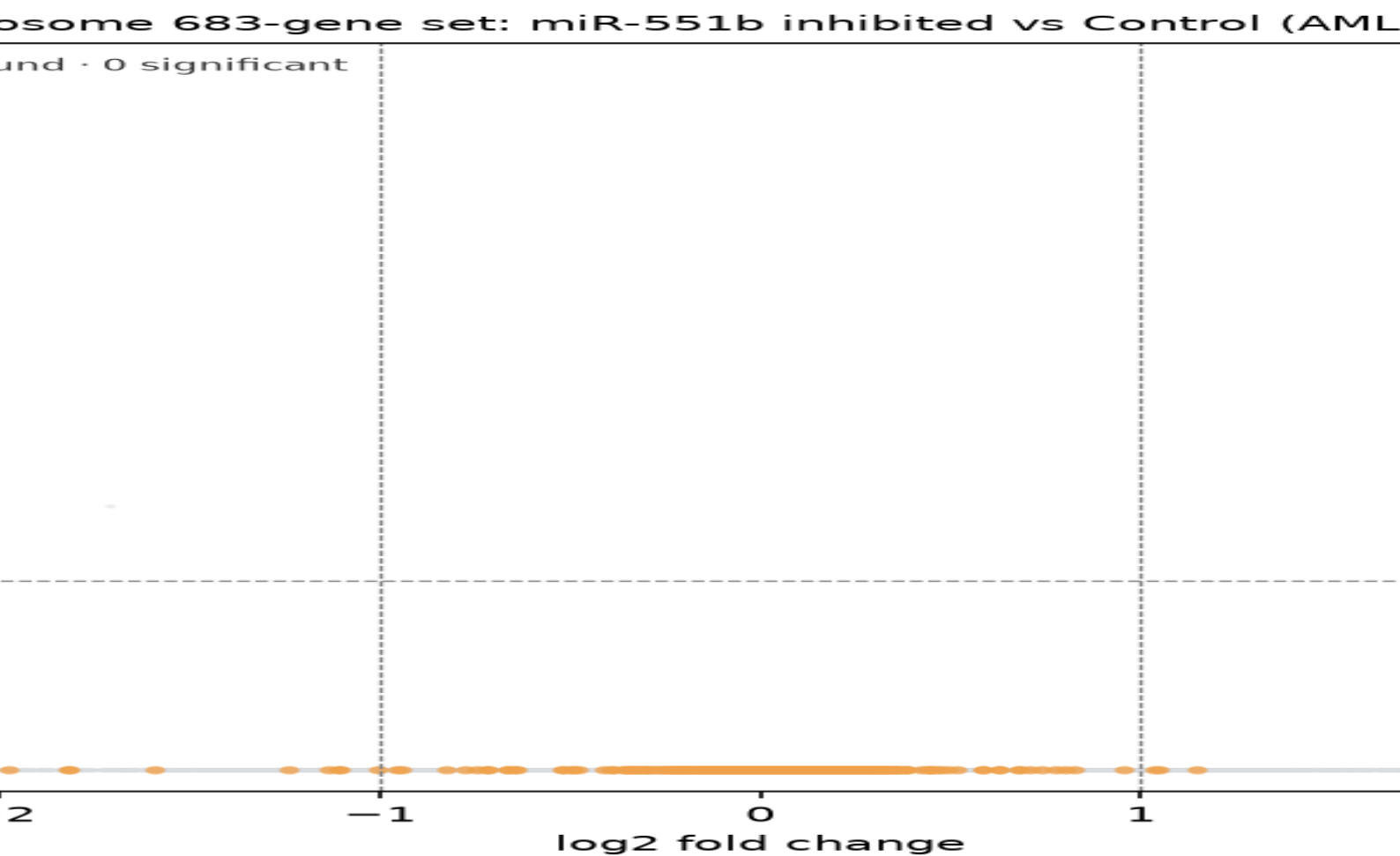


Figure 3. Volcano plot with the 683-gene cilia set overlaid (orange). No cilia gene is significant; the set sits within the unperturbed central cloud.

Table 2. Most-shifted cilia-set genes by nominal p-value (none FDR-significant).

Gene	log2FC	Nominal p	Adj. p
SYNE1	-0.52	0.012	1.00
CENPF	+0.59	0.035	1.00
HAVCR1	-1.11	0.050	1.00
RP1	+2.01	0.062	1.00
KIF14	+0.58	0.063	1.00
GSK3B	-0.28	0.082	1.00

3.4 The interaction network does not respond

Seventeen of the 25 network genes were expressed in AML; none was significant (Table 3, Figure 4). The eight non-detected members were exactly the dopaminergic module (SLC6A3, DBH, ANKK1, DRD5, PPP1R1B, DRD1, DRD2) plus IL13 - neurotransmitter genes not expressed in myeloid cells, as expected. The two ciliary members that anchored the network, IFT88 (-0.04) and ARL13B (+0.21), were flat, as were the innate-immune (TLR2/TLR4, NFKB1, JAK2), autophagy (ATG7, LAMP2, HSPA8) and iron (FTL, FTH1) members.

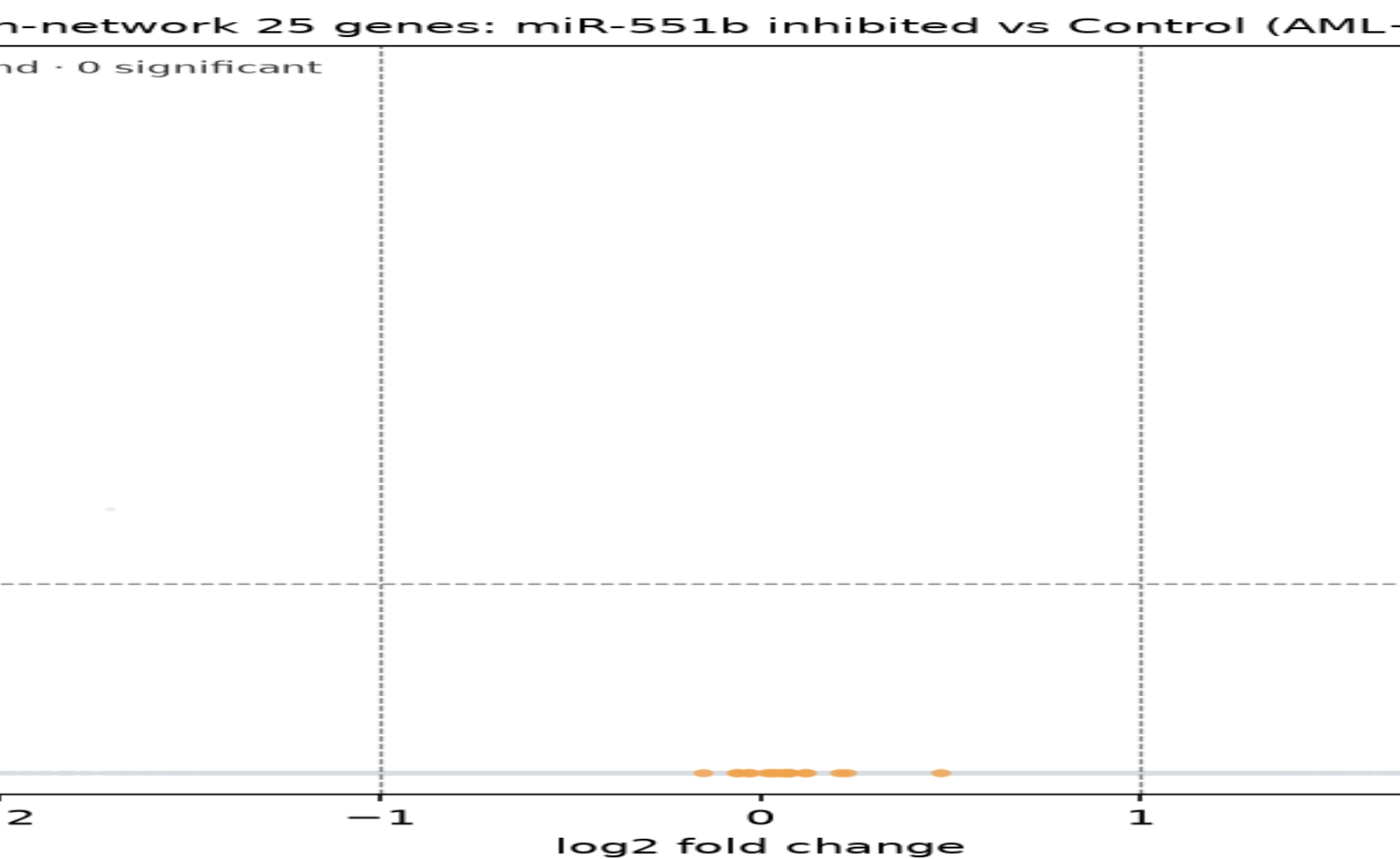


Figure 4. Volcano plot with the 25-gene interaction network overlaid (orange). All expressed members are non-significant.

Table 3. Interaction-network genes, ranked by $|\log_2\text{FC}|$ (paired model; all FDR non-significant).

Gene	log ₂ FC	Adj. p
TLR4	+0.47	~1
GYPE	+0.23	~1
ARL13B	+0.21	~1
TRAF6	-0.15	~1
HSPA8	+0.12	~1
IFT88	-0.04	~1

3.5 Per-cell-type analysis reveals a composition confound

Analysed separately, the two contexts diverge sharply. Primary AML gave 33 significant genes (8 up, 25 down), but the leading hits are overwhelmingly lymphocyte markers - TRBC1, ZAP70, LCK (T-cell), IGKC, IGHM (B-cell), plus RORA and TCF7. Because these are patient samples with variable normal immune-cell content, and each unique condition is $n=1$, this signature most plausibly reflects differing T-/B-cell contamination between the anti-miR and control samples rather than a miR-551b programme. The clean KG1a line gave only three hits, dominated by read-through/unannotated loci - i.e. essentially no reproducible effect. The pooled paired model (Section 3.2) is therefore the trustworthy read-out, and it is near-null.

4 Discussion

Across a genome-wide paired analysis, a clean-cell-line analysis, and two curated gene sets, inhibition of miR-551b left the AML transcriptome largely unchanged, and specifically did not engage ciliary/centrosomal genes - which were, if anything, more stable than average. For the cilia-in-AML hypothesis this is a clear negative: miR-551b is not an upstream regulator of the ciliary programme in these cells, and the curated interaction network is likewise unresponsive.

The one apparently strong signal - 33 genes in primary AML - is best read as a cautionary tale about cell composition in bulk RNA-seq of patient material: an immune-cell-content difference between single samples masquerades as a treatment effect. It is exactly the kind of result that replication and deconvolution, not deeper sequencing, would resolve.

Limitations. (1) The dominant limitation is replication: the 2x2x2 design provides one biological unit per unique condition, with the LNA and ZIP chemistries used as pseudo-replicates; power to detect modest effects is low. (2) miRNA inhibition often acts translationally, so a small mRNA footprint does not exclude a proteomic effect. (3) Primary-AML samples carry variable normal-cell content, confounding the per-sample contrast. (4) Quantification was alignment-free (selective alignment) rather than genome alignment. (5) Gene-set matching was by symbol; the cilia set mixes motile- and primary-cilia genes and is not myeloid-tailored. (6) No competitive set-level enrichment test was applied (with 0 hits, over-representation is uninformative).

Future directions

A replicated design (biological triplicates per condition), computational immune-cell deconvolution of the primary-AML samples, and a miR-551b target-site-aware analysis (e.g. testing predicted targets as a set) would convert this negative screen into a definitive test. If the centrosomal angle is of interest, a ciliated/centrosome-resolved model system would be more informative than bulk myeloid RNA-seq.

Data and code availability

Data: NCBI BioProject PRJNA1513000 (runs SRR40166669-SRR40166676). Reference: GENCODE v44. Analysis code: the open-source transcriptomics pipeline (github.com/imabbi47/transcriptomics); all scripts, designs and result tables for this analysis are under `cilia_aml/`.

References

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